Python-Based Linear Discriminant Analysis (LDA)

Multiple Choice Questions (MCQs) — Complete Solutions & Detailed Explanations

Quick Answer Key

Q#	Answer	Q#	Answer	Q#	Answer	Q#	Answer	Q#	Answer
1	C	2	B	3	B	4	A	5	B
6	B	7	C	8	B	9	B	10	B
11	C	12	A	13	B	14	B	15	B
16	A	17	B	18	A	19	C	21	A
22	B	23	B	24	B	25	B		

Detailed Question-by-Question Solutions

1. Which Python library provides a built-in implementation of Linear Discriminant Analysis?

A) NumPy

B) Pandas

C) Scikit-learn ✔

D) Matplotlib

> **Explanation:** `scikit-learn` contains the `LinearDiscriminantAnalysis` class in its `sklearn.discriminant_analysis` module. NumPy and Pandas are used for data structures and math operations, while Matplotlib handles plotting.

2. Which class is commonly used to implement LDA in Scikit-learn?

A) PCA

B) LinearDiscriminantAnalysis ✔

C) LogisticRegression

D) DecisionTreeClassifier

> **Explanation:** `LinearDiscriminantAnalysis` is the official class provided by Scikit-learn for fitting and transforming data using Linear Discriminant Analysis.

3. What is the purpose of the fit() method in Scikit-learn's LDA?

A) Visualize data

B) Learn discriminant directions from labeled data ✔

C) Normalize features

D) Predict labels

> **Explanation:** The `fit(X, y)` method computes the class means, total mean, and scatter matrices from labeled training data to determine the optimal linear projection vectors.

4. Which method is used to reduce dimensionality after fitting an LDA model?

A) transform() ✔

B) predict()

C) score()

D) classify()

> **Explanation:** The `transform(X)` method projects the high-dimensional feature space onto the lower-dimensional discriminant subspace computed during fitting.

5. In Python, which library is most suitable for computing eigenvalues and eigenvectors manually?

A) OpenCV

B) NumPy ✔

C) Flask

D) Seaborn

> **Explanation:** `NumPy` provides linear algebra functions, specifically `numpy.linalg.eig()` or `numpy.linalg.eigh()`, to compute eigenvalues and eigenvectors.

6. What will happen if the target labels y are not provided while fitting an LDA model?

A) LDA will automatically cluster data

B) LDA cannot be trained properly ✔

C) LDA converts to PCA

D) No effect

> **Explanation:** LDA is a *supervised* learning algorithm that relies on target labels y to compute within-class and between-class scatter. Without y, `fit()` will raise a `TypeError` or fail.

7. Which command is used to import LDA from Scikit-learn?

A) from sklearn.decomposition import LDA

B) from sklearn.lda import LDA

C) from sklearn.discriminant_analysis import LinearDiscriminantAnalysis ✔

D) from sklearn.linear_model import LDA

> **Explanation:** In modern Scikit-learn versions, the module name is `sklearn.discriminant_analysis` and the class name is `LinearDiscriminantAnalysis`.

8. What is the output shape after applying LDA with n_components = 2 to a dataset containing 100 samples?

A) (2, 100)

B) (100, 2) ✔

C) (100, 100)

D) (2, 2)

> **Explanation:** The output array preserves the number of original samples (100) while reducing the feature columns to `n_components` (2), resulting in a 2D array of shape `(100, 2)`.

9. Which NumPy function is commonly used to compute eigenvalues and eigenvectors?

A) np.mean()

B) np.linalg.eig() ✔

C) np.dot()

D) np.var()

> **Explanation:** `np.linalg.eig()` computes the eigenvalues and eigenvectors of a square array.

10. In an LDA workflow, why is feature scaling often performed before applying LDA?

A) To reduce memory usage

B) To ensure features contribute comparably ✔

C) To increase sample size

D) To remove labels

> **Explanation:** Feature scaling prevents features with larger numerical ranges/variance from dominating the scatter matrix calculations and discriminant vector direction.

11. Which parameter controls the number of discriminant components in Scikit-learn LDA?

A) n_clusters

B) n_neighbors

C) n_components ✔

D) max_depth

> **Explanation:** The parameter `n_components` specifies the number of components to keep for dimensionality reduction (bounded by $min(n_features, n_classes - 1)$).

12. Which matrix operation is required when implementing LDA from scratch?

A) Matrix inversion ✔

B) Image convolution

C) FFT

D) Hashing

> **Explanation:** To solve the generalized eigenvalue problem for LDA, one computes the eigenvalues and eigenvectors of $S_W^{-1}S_B$, which requires the inverse of the within-class scatter matrix S_W^{-1}.

13. While implementing LDA manually, which scatter matrix represents class compactness?

A) Between-Class Scatter Matrix

B) Within-Class Scatter Matrix ✔

C) Covariance Matrix

D) Correlation Matrix

> **Explanation:** The Within-Class Scatter Matrix (S_W) measures the spread/dispersion of samples around their respective class mean, representing class compactness. Minimizing S_W increases compactness.

14. Which step comes first when implementing LDA from scratch?

A) Compute eigenvectors

B) Compute class means ✔

C) Train classifier

D) Plot decision boundary

> **Explanation:** The foundational step in scratch LDA is calculating the mean vector for each class (and overall mean), which is required to calculate the scatter matrices.

15. After computing eigenvectors in manual LDA, what should be done next?

A) Randomly shuffle them

B) Sort them according to eigenvalues ✔

C) Delete negative values

D) Normalize labels

> **Explanation:** Eigenvectors must be sorted in descending order based on their corresponding eigenvalues so that the vectors capturing the maximum class separation are selected first.

16. Which NumPy function can be used to project data onto selected discriminant vectors?

A) np.dot() ✔

B) np.max()

C) np.sum()

D) np.argmax()

> **Explanation:** The projection is performed via matrix multiplication between the original data matrix X and projection matrix W: `X_lda = np.dot(X, W)` or `X.dot(W)`.

17. What is the primary purpose of lda.predict(X_test)?

A) Reduce dimensions

B) Classify unseen samples ✔

C) Scale data

D) Compute covariance

> **Explanation:** `lda.predict(X_test)` uses the learned Bayes decision boundaries to assign categorical class labels to new test instances.

18. Which metric is commonly used to evaluate an LDA classifier?

A) Accuracy ✔

B) RMSE

C) MAPE

D) MAE

> **Explanation:** LDA is primarily used for classification, making Accuracy (or Precision, Recall, F1-Score) the standard evaluation metric. RMSE, MAPE, and MAE are regression metrics.

19. In Scikit-learn, LDA can be used for:

A) Classification only

B) Dimensionality reduction only

C) Both classification and dimensionality reduction ✔

D) Clustering only

> **Explanation:** LDA projects data into a lower-dimensional space (dimensionality reduction) while simultaneously finding linear decision boundaries to categorize data (classification).

21. You are asked to implement Linear Discriminant Analysis from scratch using only NumPy... Which sequence of steps is correct?

A) Standardize Data → Compute Class Means → Compute Within-Class Scatter Matrix → Compute Between-Class Scatter Matrix → Solve Eigenvalue Problem of $S_W^{-1}S_B$ → Select Top Eigenvectors → Project Data ✔

B) Compute FFT → Compute Covariance Matrix → Apply K-Means → Project Data

C) Compute PCA Components → Apply Logistic Regression → Predict Labels

D) Normalize Labels → Compute Decision Tree → Extract Features → Project Data

> **Explanation:** Option A describes the standard mathematical pipeline for scratch LDA implementation.

22. What is the purpose of the following line in a manual LDA implementation?

```
eigvals, eigvecs = np.linalg.eig(np.linalg.inv(Sw).dot(Sb))
```

A) Compute principal components

B) Compute discriminant directions ✔

C) Compute cluster centroids

D) Compute feature scaling factors

> **Explanation:** Solving $S_W^{-1}S_B v = \lambda v$ yields eigenvectors (v) that represent the directions (linear discriminants) maximizing between-class variance relative to within-class variance.

23. Consider: `X_lda = X.dot(W)`. What does W represent?

A) Weight initialization matrix

B) Matrix of selected discriminant vectors ✔

C) Covariance matrix

D) Distance matrix

> **Explanation:** W is the projection matrix formed by stacking the top k selected linear discriminant eigenvectors as columns.

24. In a scratch implementation, why are eigenvectors sorted in descending order of eigenvalues?

A) To reduce execution time

B) To select directions with maximum discriminative power ✔

C) To normalize data

D) To balance classes

> **Explanation:** Larger eigenvalues correspond to eigenvectors that achieve greater separation between classes. Sorting in descending order ensures we pick the axes that retain maximum class separability.

25. When implementing LDA manually, what is the most likely issue if Sw is singular?

A) Labels disappear

B) Matrix inversion fails ✔

C) Data becomes nonlinear

D) Eigenvalues become complex automatically

> **Explanation:** If S_W (Within-Class Scatter Matrix) is singular (non-invertible, which happens when sample size < number of features or features are perfectly collinear), `np.linalg.inv(Sw)` will raise a `LinAlgError`.